

Routledge
Taylor & Francis Group

International Journal of Disability, Development and Education

ISSN: 1034-912X (Print) 1465-346X (Online) Journal homepage: www.tandfonline.com/journals/cijd20

A Comparison of Animated and Adult Video Modelling in Teaching Social-Communication Skills to 3-Year-Olds with Autism Spectrum Disorders

Huan Li, Yan Peng, Zhaojun Li, Jiaying Lin, Hongyu Wu, Judith J. Carta, Xiangyu Li & Mengxi Wang

To cite this article: Huan Li, Yan Peng, Zhaojun Li, Jiaying Lin, Hongyu Wu, Judith J. Carta, Xiangyu Li & Mengxi Wang (2025) A Comparison of Animated and Adult Video Modelling in Teaching Social-Communication Skills to 3-Year-Olds with Autism Spectrum Disorders, International Journal of Disability, Development and Education, 72:2, 213-234, DOI: 10.1080/1034912X.2024.2355339

To link to this article: https://doi.org/10.1080/1034912X.2024.2355339

Published online: 20 May 2024.

Submit your article to this journal

Article views: 431

View related articles

View Crossmark data

Full Terms & Conditions of access and use can be found at
https://www.tandfonline.com/action/journalInformation?journalCode=cijd20

INTERNATIONAL JOURNAL OF DISABILITY, DEVELOPMENT AND EDUCATION
2025, VOL. 72, NO. 2, 213–234
https://doi.org/10.1080/1034912X.2024.2355339

Routledge
Taylor & Francis Group

Check for updates

A Comparison of Animated and Adult Video Modelling in Teaching Social-Communication Skills to 3-Year-Olds with Autism Spectrum Disorders

Huan Li[a,b], Yan Peng[c], Zhaojun Li[a], Jiaying Lin[a], Hongyu Wu[a], Judith J. Carta[b], Xiangyu Li[d] and Mengxi Wang[a]

[a]Department of Special Education, Southwest University, Chongqing, China; [b]Juniper Gardens Children's Project, University of Kansas, Kansas City, USA; [c]Special Education, Shenzhen Nanshan Longyuan School, Guangdong, China; [d]Faculty of Education, University of Macau, China

ABSTRACT

This study aims to compare the effectiveness of emerging animated video modelling (VM) and the most widely used adult VM in teaching social-communication skills to 3-year-olds with autism spectrum disorders (ASD) in China. A single-case experimental design was used, with data analysed by using visual analysis and an effect size measure, namely, the baseline corrected Tau (Tau-BC). Three 3-year-old preschoolers participated in this study. Each preschooler's social-communication behaviours across the baseline, intervention, withdrawal, and maintenance conditions were recorded and analysed. Results showed that all three VM effectively improved preschoolers' social-communication behaviours; preschoolers' improvement in overall social communication and scripted social communication was greater in response to animated VM than adult VM; two of the three preschoolers' improvement in unscripted social communication was greater in response to animated VM than adult VM. Mixed VM led to the greatest improvement in all three social-communication behaviours for two of the three preschoolers. The preliminary findings extended previous findings by demonstrating for the first time that animated VM was more effective than adult VM, and mixed VM was more effective than animated VM or adult VM alone for two of the three preschoolers.

ARTICLE HISTORY
Received 3 June 2021
Accepted 30 November 2021

KEYWORDS
Video modelling; autism; animation; social communication; scripts; preschoolers

Introduction

Social-communication skills refer to a range of verbal and nonverbal behaviours used in reciprocal social interaction (Wetherby et al., 2007). These skills, as critical milestones in the development of children, are strongly associated with long-term outcomes (Howlin et al., 2004). Researchers have long reported deficits in social-communication skills as the earliest identifiable feature of autism spectrum disorders (ASD; Ingersoll & Wainer, 2013). Deficits in these skills can negatively impact children's management of everyday interactions with others, their development and maintenance of meaningful peer relationships, and their inclusion opportunities in both school and community settings (Koegel et al.,

CONTACT Huan Li lihuanswu@gmail.com
© 2024 Informa UK Limited, trading as Taylor & Francis Group

2013). Thus, it is critical to improve social-communication skills in early development for children with ASD (Ingersoll, 2011).

Video modelling (VM) is an evidence-based intervention strategy for teaching social-communication skills to children with ASD (Hume et al., 2021) and is theoretically rooted in social learning theory (Bellini & Akullian, 2007). This theory underscores that human behaviour is primarily learned by observing and modelling others, and thus VM generally involves the subject observing a video of a model engaging in a behaviour that is subsequently practised and imitated (Corbett & Abdullah, 2005).

Types of Video Modelling

Early uses of VM for teaching social-communication skills to children with ASD involved human demonstrating VM, including adult VM, self VM, peer VM, and point-of-view VM (Shukla-Mehta et al., 2010). The most commonly used among these variations was adult VM (Bellini & Akullian, 2007; Mason et al., 2012). A growing number of systematic reviews have supported the effectiveness of human demonstrating VM to improve the social-communication skills of children with ASD (Mason et al., 2013; A. McCoy et al., 2016; Qi et al., 2018).

More recently, animated VM, an emerging VM intervention that incorporates animation, has been found effective in instruction for children with ASD. Studies have found that animated VM is a rapid and effective method for toilet training (Drysdale et al., 2015), social skills (Charlton et al., 2020), joint attention and social engagement (Ho et al., 2019), and daily living skills (Ohtake et al., 2015). Neuroscience research has laid the foundation for the effectiveness of animated VM (Grelotti et al., 2005; Rosset et al., 2008). Studies in individuals with ASD have consistently demonstrated abnormal hypoactivation in the amygdala and fusiform gyrus area, brain areas that participate in human face processing. However, individuals with ASD hyperactivated these areas when extracting information from cartoon characters, and the processing in cartoon characters was faster than human faces and objects (Grelotti et al., 2005). In addition, a study that used an approach-avoidance task indicated that individuals with ASD showed more quick avoidance from human photographs than cartoons and greater approach to cartoons than human photographs (Silva et al., 2015). Based on the above evidence, we assumed that animated VM might improve social communication for children with ASD more effectively than human demonstrating VM. However, no study has compared the effects of animated VM and other VM, and the number of studies that explore the effects of animated VM in teaching social communication for children with ASD is very limited. A possible reason for the limited number of studies of animated VM might be that knowledge about simple and time-efficient approaches for producing animated videos has not been widely available. This study introduces a simple and economical way to produce animated videos.

Furthermore, researchers have also tried to explore whether mixed VM (i.e. combining two or more types of VM within one intervention session) could augment the effects of a single type of VM in teaching social-communication skills to children with ASD. However, the number of studies on mixed VM is limited and these studies have yielded inconsistent results in teaching children with ASD. Maione and Mirenda (2006) found that a mixed VM approach combining adult VM and VM feedback augmented the effect of adult VM alone in increasing the social language of children with ASD, while Nikopoulos

and Keenan (2003) reported limited success in three out of seven participants for whom adult, peer, and self-VM were utilised. Based on the review of studies on the mixed VM, Mccoy and Hermansen (2007) proposed that children might have different preferences for the types of VM and the mixed VM might increase the possibility of meeting the needs of different children. Meanwhile, different combination forms of mixed MV may produce inconsistent results. Thus, this study attempted to explore whether mixed MV which combined the most commonly used VM (adult VM) and the emergent animated VM was more effective on the social communication skills of children with ASD than a single type of VM.

Comparison of the Effectiveness of Different Video Modelling

As an increasing number of studies have demonstrated the effectiveness of different VMs for the social communication of children with ASD, researchers have begun to focus on comparing the effectiveness of different VMs. However, the number of comparative studies of different VMs for improving social communication is limited, and the comparison has yielded inconsistent results. Sherer et al. (2001) found that peer and self VM were equally effective for increasing conversation skills in children with ASD, while Marcus and Wilder (2009) found that self VM was more effective and efficient than peer VM in teaching children with ASD to respond appropriately. The aforementioned evidence may indicate that more studies are needed to obtain more convincing results in comparing the effects of different VMs. However, no study has compared the effectiveness of animated VM and human demonstrating VM.

Research Questions

The current study sought to address three major gaps in the literature: (1) the very limited number of studies that explore the effectiveness of animated VM in improving social-communication skills for preschoolers with ASD; (2) the insufficient evidence of the effectiveness of mixed VM; and (3) the absence of research that compares the effectiveness of animated VM, human demonstrating VM, and mixed VM. Therefore, we set out to answer two major research questions. (1) Would animated VM or adult VM (i.e. the most commonly used human demonstrating VM) be more effective in teaching Chinese preschoolers with ASD social-communication skills? (2) Would the mixed VM that combines animated VM and adult VM be more effective than the two VMs alone?

Methods

Participants

In this study, three preschoolers with ASD were recruited from a private inclusive preschool in a western city in China. Through observations and conferencing with classroom teachers, three preschoolers were enrolled in the study based on the following inclusion criteria: (1) a diagnosis record of ASD by physicians or clinical psychologists; (2) the ability to imitate simple gross motor movements, such as touching the head and clapping hands (Charlop-Christy et al., 2000; Shipley-Benamou et al., 2002); (3) the ability to watch and

sustain attention to a video for at least 5 min (Charlop-Christy et al., 2000; Nikopoulos & Keenan, 2003); (4) parental consent for participation; and (5) no previous exposure to VM.

Anan

Anan was a boy aged 3 years and 0 months, and his IQ records from the Chinese version of the Wechsler Preschool and Primary Scale of Intelligence, 4th edition (WPPSI-IV; Wechsler, 2012) were as follows: Full-Scale IQ (FSIQ) – 65; Verbal Comprehension (VCI) – 57; Visual-Spatial (VSI) – 70; and Working Memory (WMI) – 63. The medical records of the Autism Diagnostic Observation Scale (ADOS) showed that Anan scored within the autism range for communication, social interaction, and total score. Anan was observed to imitate simple words and understand simple commands but seldom communicated with others, rarely looked at and responded to others, and often used nonverbal communication behaviours, such as dragging and pointing.

Bobo

Bobo was a boy aged 3 years and 11 months, and his IQ records from the WPPSI-IV were FSIQ – 85, VCI – 77, VSI – 90, and WMI – 83. The ADOS records showed that he scored within the autism range for communication, social interaction, and total score. Bobo was observed to communicate by using simple words or actions to express demands and understood commands. He presented obvious pronunciation and intonation problems and severe emotional behaviour problems, e.g. repeated and unusual vocalisation and yelling or shouting at others, when his demands were not met.

Cici

Cici was a girl aged 3 years and 10 months, and her IQ records from the WPPSI-IV were FSIQ – 60, VCI – 51, VSI – 65, and WMI – 63. The ADOS records showed that she scored within the autism range for communication, social interaction, and total score. Cici was observed to frequently engage in perseverative speech and self-stimulatory behaviours and presented excessive adherence to routines and poor concentration.

Study Design

To compare the effectiveness of three types of VM for teaching social-communication skills to three preschoolers with ASD in China, this research applied a combined single-case experimental design, including a multiple-baseline design, alternating treatments design, and multitreatment design (Wolery et al., 2018). Thus, this study consisted of seven phases, namely, baseline (A), the first alternating treatments phase (B), the first mixed VM phase (C), withdrawal (A), the second mixed VM phase (C), the second alternating treatments phase (B), and maintenance (A), i.e. an A-B-C-A-C-B-A design.

The multiple-baseline design was used to explore the effectiveness of video modelling across children. Additionally, the primary goal of this study is to compare the effectiveness of animated VM and adult VM in improving social-communication skills, and thus only these two VM interventions were compared in the alternating treatments design to demonstrate their relative effects more clearly. In the alternating treatments phases (phase B), animated VM and adult VM were implemented on alternate days, and the order of the two VM interventions was presented randomly across children, to eliminate

the sequence effect. Finally, the multitreatment design was used to compare the effectiveness of mixed VM (phase C) and a single use of animated VM or adult VM (phase B). The order of phases B and C was also exchanged after withdrawal, to eliminate the sequence effect. This study was conducted with the approval of the Research Ethics Committee of the first author's university.

Settings and Materials

All sessions were conducted in a LEGO® activity room ($3 \times 4 \times 3$ m) at the participants' preschool. Preliminary observations and interviews with the preschoolers' parents and teachers showed that the three preschoolers were interested in colourful operational materials and preferred building blocks. Therefore, we chose LEGO® blocks as teaching materials. Larger-sized LEGO® blocks for children from 2–6 years, 10 pictures of LEGO® models, a video camera, and a timer were present in the setting throughout the study. These LEGO® models were recommended by the teachers of the children, including a car, train, sofa, tree, dragonfly, airplane, truck, house, rocket, and scarecrow.

Scripts

Similar to other studies that used VM to teach social-communication skills (Boudreau & D'Entremont, 2010; Maione & Mirenda, 2006), this study developed 10 scripts that targeted social-communication skills (listed in Table 1). To ensure that the scripts were based on the children's communication needs and were of an appropriate length and complexity to address the children's expressive language levels (Ganz, 2007), we interviewed the children's special education teachers and parents. In addition, to ensure that the scripts were age-appropriate, we analysed the social interactions of typical children playing with LEGO® blocks.

Table 1. Social-communication scripts in the videos.

	Scripts of CP	Types of scripts	Target scripts of participants
1	What's this?	Social response	Verbal: This is a car (the LEGO® model in the picture). Nonverbal: Pointing to the picture of the LEGO® model.
2	Which colour of blocks do you like?	Social response	Verbal: I like red (colour) blocks. Nonverbal: Pointing to the red (colour) block.
3	Shall we play together?	Social response	Verbal: Ok, let's play. Nonverbal: Taking blocks with hands.
4	Can you give that block to me?	Social response	Verbal: Ok, here you are. Nonverbal: Passing the blocks to the CP.
5	Where are you going to put this block?	Social response	Verbal: Put it next to (locality) the red (colour) block. Nonverbal: Pointing to the right position as the LEGO® model.
6	Ok, let's play.	Social initiation	Verbal: Shall we play together? Nonverbal: Showing a block.
7	Ok, I'll help you.	Social initiation	Verbal: Can you help me? Nonverbal: Pointing to the blocks.
8	Ok, here you are.	Social initiation	Verbal: Please give me the yellow (colour) block. Nonverbal: Pointing to the yellow block.
9	Thanks for your reminder.	Social initiation	Verbal: This blue (colour) block is misplaced. Nonverbal: Taking away the wrong block.
10	Wow, good job!	Social initiation	Verbal: Look, I made a car (the final product)! Nonverbal: Pointing to the blocks and high fiving the CP.

Note: CP = communication partner. Words with parentheses after them indicate that these words may vary in different videos. Taking the No. 1 target script as an example, the verbal response script could be 'This is a sofa/house/tree . . . ' in other videos.



Figure 1. A screenshot of videos with adult demonstrations.

Videos with Animated and Adult Demonstrations

There were two categories of videos in this study: videos with adults carrying out the demonstrations (see Figure 1) and videos with animated characters carrying out the demonstrations (see Figure 2). In the videos with adult demonstrations, the second author played the role of the communication partner (CP), and a graduate student majoring in special education played the role of the child. In the animated videos, two animated bunnies demonstrated the same scripts as the videos with adult demonstrations. Since each video demonstrated one communication script with one specific LEGO® model and there were 10 scripts and 10 LEGO® models, we produced 100 videos with adult demonstrations and 100 animated videos. Each video lasted for 15 to 20 s.



Figure 2. A screenshot of animated videos.

We produced the videos with adult demonstrations by video-recording the two researchers demonstrating each communication script using LEGO® activities. We produced animated videos by using the FaceU app on iPad to video-record the two researchers demonstrating the same communication scripts using LEGO® activities. FaceU allows users to substitute animated faces for human faces with one click. In the animated videos, animated bunny faces were chosen to replace human faces according to the teachers' and parents' suggestions.

Independent Variables

This study included three independent variables: animated VM, adult VM, and mixed VM. Animated VM was defined as VM instruction delivered to children using animated videos. Adult VM was defined as VM instruction delivered to children using videos with adult demonstrations. Mixed VM was defined as VM instruction delivered via animated videos and videos with adult demonstrations within one intervention session. All three interventions taught children to master social-communication skills.

Dependent Variables

In this study, three social-communication skills were measured as the dependent variables, i.e. the frequency of scripted social communication (including scripted verbal communication and scripted nonverbal communication), the frequency of unscripted social communication (including unscripted verbal communication and unscripted nonverbal communication), and the frequency of overall social communication (i.e. the total frequency of scripted and unscripted communication). The definitions of these dependent variables, presented in Table 2, were adapted from related studies (Topuz & Ulke-Kurkcuoglu, 2019; Wichnick-Gillis et al., 2016).

Table 2. Definitions of social-communication outcomes.

Type	Subtype	Definition
Scripted social communication	Verbal communication	A verbal production matching a script in video modelling (VM), including an initiation or a response to the communication partner (CP). It does not have to be grammatically correct. A production with changes in conjunctions, prepositions, particles, and interjection words is still scored as scripted (Krantz & McClannahan, 1998).
	Nonverbal communication	A nonverbal production matching a script in VM, also including an initiation or a response.
Unscripted social communication	Verbal communication	A verbal production different from the script in content words, including nouns, verbs, adjectives, adverbs, quantifiers, and pronouns; it needs to be contextually relevant to the given activity and phonologically and semantically understandable. Interactions do not have to be grammatically correct.
	Nonverbal communication	A nonverbal production that differed from the script in the VM but contextually relevant to the current activity. Nonverbal interactions are defined as 9 forms based on a previous study (Stone et al., 1997), including Give Objects, Show Objects, Touch Objects, Manipulate the Partner's Hand, Reach, Contact Point, Distal Point, Vocalization and Eye Gaze towards the CP.

Measurement

We developed a 10-min LEGO® activity to assess the social-communication outcomes of children in each session of baseline, intervention, withdrawal, and maintenance. To avoid the priming effect (Bargh & Chartrand, 2000), the participants returned to the regular curriculum in their classroom after each intervention session, and 30 min later, they returned to the LEGO® activity classroom for assessment. In the assessment, the CP played with the LEGO® blocks with the children and provided them with opportunities to demonstrate their social-communication skills. Each assessment included four trials, and each trial had a different LEGO® model to provide more opportunities for the children to demonstrate their skills and examine the children's generalisation in different activity materials. Each trial lasted approximately 2–3 min, which made the entire assessment activity 10 min.

The procedures of each trial were as follows. (1) Model selection. The model was randomly selected by the CP in the first trial, and the children selected models in the remaining three trials. (2) Drawing attention. The CP presented the picture of the LEGO® model and drew the children's attention to the model by saying, 'Now, we'll make the XXX together'. (3) Expecting the initiation of children. The CP intentionally provided an initiation opportunity for participants and expected the child to initiate a conversation regarding the activity. In the intervention sessions, the CP provided an opportunity for emitting the initiation script in the video; in the baseline, withdrawal, and maintenance sessions, the CP randomly provided an opportunity for one of the initiation scripts. The CP responded appropriately when the children initiated a conversation verbally or nonverbally. If the children did not initiate any interaction after being provided with the initiation opportunity, then the CP played with the blocks with the children without any verbal response. If the child did not pick up a block after the play began, then the CP put a block in the child's hand and guided the child by repeating 'Let's make the XXX'. (4) Expecting the response of children. The CP emitted an initiation script and expected the child to respond correctly. In the intervention sessions, the CP emitted the initiation script in the video; in the baseline, withdrawal, and maintenance sessions, the CP randomly emitted one of the initiation scripts. No other instructions, prompts, or reinforcement were given.

We video-recorded the 10-min LEGO® assessment activities within all sessions for the purpose of data collection. Two researchers recorded the frequencies of the three dependent variables by watching the video recordings (i.e. by counting the total number of scripted communications, unscripted communications, and overall social communication in each assessment activity). In addition to the definition of the three social-communication outcomes, some additional measurement rules were as follows. (1) The researchers recorded the frequencies of the dependent variables regardless of their length. Specifically, a single utterance that could express an independent statement, question, request, command, exclamation, etc., was also counted. (2) The nonrelevant interactions were excluded. For example, if the CP asked the child, 'What's this?' and the child answered, 'I'm Anan', then this interaction would not be counted. (3) Delayed echolalia was also excluded, such as 'I made a car; I made a car; I made a car; I made a car'. (4) If the child uttered the CP's script, it was considered to be a scripted verbal interaction.

Procedures

Baseline

We examined the baseline performance of the social-communication behaviours of children in a 10-minute LEGO® assessment activity. The criterion for moving from baseline to intervention was stable responding for three consecutive sessions. Specifically, following a stable baseline performance (i.e. a minimum of three consecutive data points falling within a 20% range and no increasing trend across data points) in the frequency of overall social communication, the intervention was introduced to the first child; when the first child's three consecutive data points in the intervention showed no overlap with the baseline, we introduced the intervention to the second child. The same criterion was used to determine when to introduce the intervention to the third child.

Intervention

Two researchers conducted the VM interventions. The second author, who was also the CP in the videos, communicated and played with the children in the LEGO® activity. The other research assistant sat behind the children and provided prompts when needed. The VM interventions were implemented individually every school day morning. Each intervention session lasted approximately 20 min.

The procedures of each intervention session are presented in Figure 3. In each intervention session, the CP randomly selected a video of a social response script and a video of a social initiation script (see Table 1) as the target videos. The VM intervention that focused on the response script was conducted before the intervention on the initiation script. The intervention for each script followed the same procedures and consisted of three rounds of VM and practice. The procedures in adult VM, animated VM, and mixed VM were exactly the same, except for the different types of videos used in the interventions. The following is a detailed introduction to the intervention procedures of the response script.

The first round of VM with full repeating. (1) In the animated VM and adult VM sessions, each video of the target response script was played two consecutive times; in the mixed VM sessions, a video with adult demonstrations and an animated video of the same target script were played once. (2) The CP presented the same picture of the LEGO® model as the video and drew the child's attention by saying, 'Now, we'll make this model together'. (3) After playing with the blocks for 30 s, the CP initiated the CP's script in the video and waited for the child's response. If the child did not respond correctly within 5 s, then the assistant modelled the first word of the targeted response script. If the child still did not respond appropriately, then the assistant modelled the entire targeted script once and asked the child to repeat the response. (4) Differential reinforcement was used for the prompted and unprompted responses of the child. If the child emitted the script correctly without prompts, then the CP reinforced the child enthusiastically by caressing the child's head and saying, 'Well done!' in a higher tone. The CP reinforced the child in a less enthusiastic way for prompted correct responses (Topuz & Ulke-Kurkcuoglu, 2019). If the child interacted using an unscripted and appropriate response without prompts, then the CP also reinforced the child enthusiastically and responded to the child appropriately.

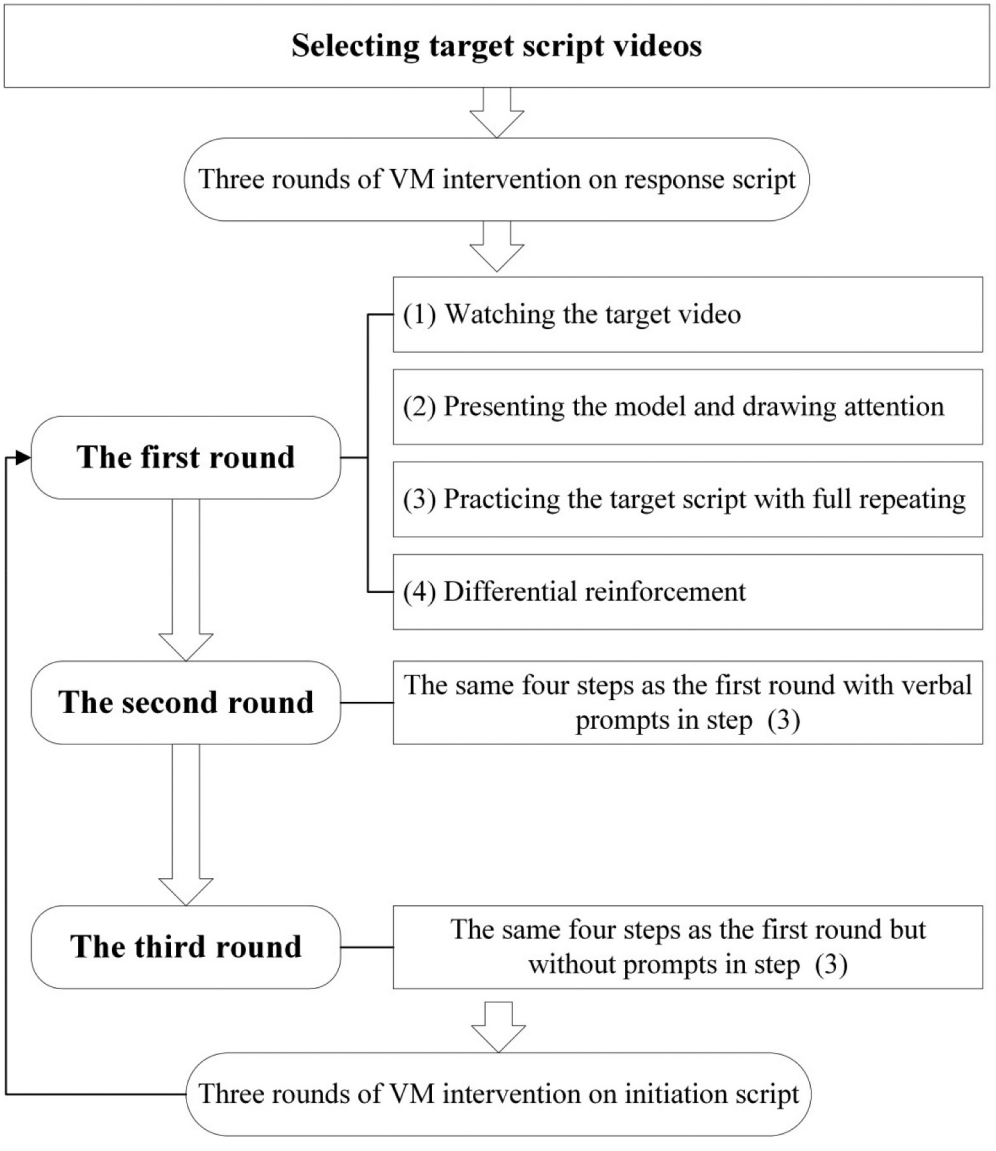


Figure 3. Flowchart of VM Interventions in Each Session. Note. VM = video modelling

The second round of VM with verbal prompts. The second round of intervention procedures for the response script repeated the four steps in the first round. However, in step (3), if the child did not respond correctly, then the assistant only provided verbal prompts by modelling the first word of the targeted response script, without repeating the full script.

The third round of VM without prompts. The third round of intervention procedures for the response script repeated the four steps in the first round but without any prompt in step (3). After the completion of the intervention that focused on the response script, we conducted another three rounds of instruction that focused on the initiation script.

Withdrawal and Maintenance

Withdrawal and maintenance data on social-communication outcomes were collected during the 10-min LEGO® assessment, consistent with the baseline and intervention conditions. Specifically, in the withdrawal condition, three data points were collected when withdrawing the VM intervention for each participant. According to Rusch and Kazdin's (1981) study, although the demonstration that the total withdrawal of the intervention results in a contra-therapeutic trend reveals that a functional relationship exists between treatment and behavioural change, this return to baseline implies that efforts to maintain treatment gains would need to be programmed. Thus, in the maintenance condition, a partial-withdrawal strategy was used (Rusch & Kazdin, 1981). Specifically, the participant watched a randomly selected video of a social response script and a video of a social initiation script two consecutive times, but the instruction, prompts, and feedback were withdrawn. Three maintenance data points were collected for each child at one week, two weeks, and one month after the final intervention session.

Interobserver Agreement

The second author trained a research assistant who was blind to the research to code the dependent variables from the training videos. After the second author and the assistant reached 90% reliability, the second author coded all videos, and the assistant randomly selected and recoded videos for approximately 30% of the sessions of each condition and each child. Interobserver agreement was calculated by a general comparison of frequency counts in which the number of agreements was divided by the number of agreements plus disagreements and multiplied by 100 (Ayres & Gast, 2010). The average interobserver agreement across conditions for Anan, Bobo, and Cici was 92.56%, 98.51%, and 91.07%, respectively.

Procedural Fidelity

Procedural fidelity was calculated by dividing the number of correctly completed researcher behaviours by the total number of expected researcher behaviours and multiplying by 100 (Gast, 2010). Two independent assistants randomly selected and assessed 30% of the sessions at baseline, intervention, withdrawal, and maintenance to evaluate whether the steps were implemented with reliability based on the procedure fidelity checklist. The average procedural fidelity across the participants for baseline, intervention, withdrawal, and maintenance was 99.20%, 93.25%, 99.50%, and 99.50%, respectively; the mean average procedural fidelity across conditions for Anan, Bobo, and Cici was 98.15%, 98.14%, and 94.44%, respectively. The agreement on procedural fidelity by the two assistants was 95.35%.

Social Validity

In the final stage of the study, the children's teachers and parents completed a questionnaire about the satisfaction and acceptability of the three VM interventions. All of them indicated satisfaction with all VM interventions. Particularly, all parents and

teachers responded that animated VM was a good teaching strategy and they would consider using this intervention to teach their children, and they asked for more resources of animated VM.

Data Analysis

Visual analysis and an effect size measure, the Baseline Corrected Tau (Tau-BC), were used to evaluate the effects of the VM interventions. Specifically, we visually analysed the dependent variables for the level, trend, variability, overlap, immediacy of effect, and consistency of data patterns. We compared the levels of the dependent variables across each condition by the means and ranges of the values, and trends in the data were calculated by using the split middle method technique (Ledford & Gast, 2018). The Tau-BC is an effect size statistic based on rank correlation and robust regression, which purports to control for baseline trend (Tarlow, 2017), and it is calculated by a web-based calculator (Tarlow, 2017). The intervention with a higher level of the Tau-BC indicates that the method is superior.

Results

Overall Social Communication

For all children, the frequency of overall social communication immediately increased to a relatively high level with a gradually increasing trend at the beginning of the intervention conditions (see Figure 4). The visual analysis showed no overlap between the baseline and the intervention conditions or between the baseline and maintenance conditions for Anan and Cici, and Bobo's data also demonstrated the same pattern, except for the first data point in the adult VM condition, i.e. a 3.33% overlap between baseline and intervention.

Comparison of Animated VM and Adult VM

Figure 4 reveals that Anan and Bobo demonstrate an obviously higher level of overall social-communication frequency in animated VM compared to the adjacent adult VM conditions. Cici's results repeat this pattern, although the difference is slight. Specifically, all children showed a higher frequency of overall social communication in response to the first animated VM compared to the first adult VM, and this pattern was repeated in the second animated VM compared to the second adult VM. The Tau-BC results were consistent with the visual analysis (see Table 3).

Comparison of Mixed VM and Single VM

Anan's and Bobo's level of overall social-communication frequency in the mixed VM conditions was higher than that in the adjacent adult VM and animated VM conditions (see Table 4). Although Cici's level of overall social-communication frequency in the second mixed VM was lower than that of the adult VM and animated VM conditions, the Tau-BC results reveal that mixed VM produces higher effects in overall social communication than animated or adult VM for all children.

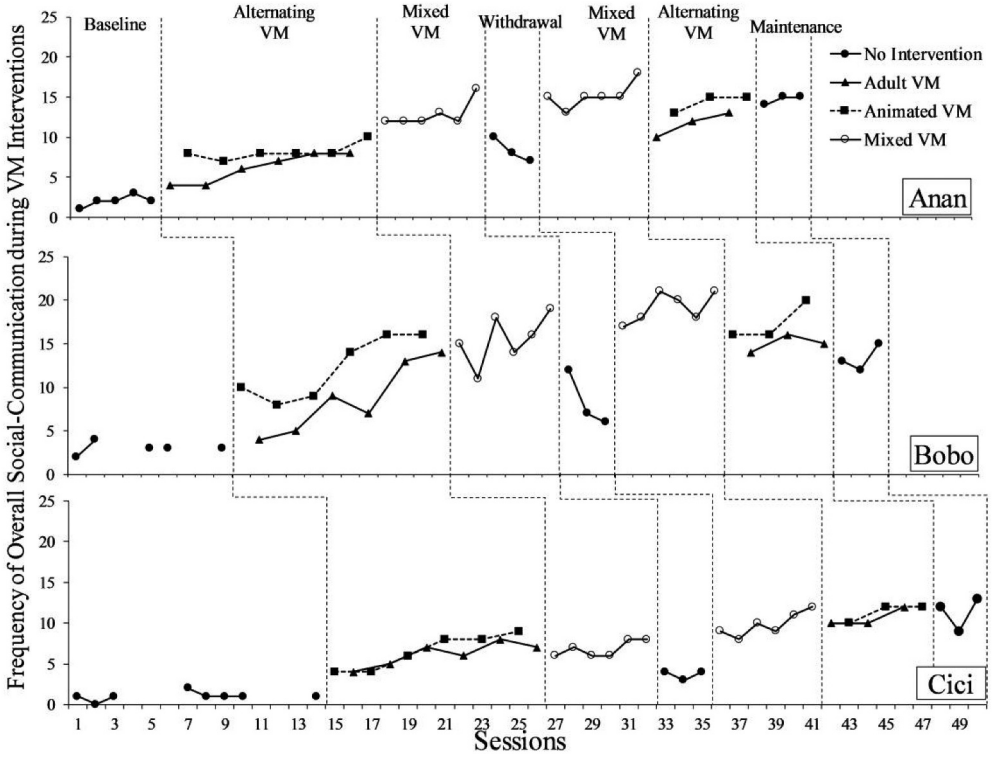


Figure 4. Frequency of overall social-communication during VM Interventions. Note. VM = video modelling

Table 3. Baseline corrected tau of VM interventions for participants.

	Overall social-communication			Scripted social-communication			Unscripted social-communication		
Participants	Adult VM	Animated VM	Mixed VM	Adult VM	Animated VM	Mixed VM	Adult VM	Animated VM	Mixed VM
Anan	0.723	0.745	0.808	0.741	0.756	0.808	0.556	0.808	0.816
Bobo	0.707	0.741	0.775	0.544	0.688	0.799	0.752	0.767	0.835
Cici	0.778	0.781	0.800	0.788	0.808	0.828	0.844	0.770	0.808

Note: VM = video modelling.

During the withdrawal conditions, the total withdrawal of the intervention resulted in a significantly lower level of overall social-communication frequency with a decreasing trend. However, in the maintenance conditions with partial withdrawal of the intervention, all children's overall social-communication data remained at a high level similar to that of the intervention. This differentiated result might reveal that the intervention component maintained in the maintenance conditions, i.e. the independent video viewing, was responsible for maintenance.

Table 4. Means and ranges of frequencies of overall social-communication, scripted social-communication, and unscripted social-communication for participants.

Participants	Baseline	The first animated VM	The first adult VM	The first mixed VM	Withdrawal	The second animated VM	The second adult VM	The second mixed VM	Maintenance
				Overall social-communication					
Anan	2.00(1–3)	8.16(7–10)	6.17(4–8)	12.83(12–16)	8.33(7–10)	14.33(13–15)	11.67(10–13)	15.17(13–18)	14.67(14–15)
Bobo	3.00(2–4)	12.17(8–16)	8.67(4–14)	15.50(11–19)	8.33(6–12)	17.33(16–20)	15.00(14–16)	19.17(17–21)	13.33(12–15)
Cici	1.00(0–2)	6.50(4–9)	6.17(4–8)	6.83(6–8)	3.67(3–4)	11.33(10–12)	10.67(10–12)	9.83(8–12)	11.33(9–13)
				Scripted social-communication					
Anan	1.80(1–2)	5.50(4–7)	5.00(4–6)	9.33(8–11)	7.00(6–8)	11.00(10–12)	9.00(8–10)	11.33(10–13)	11.33(10–12)
Bobo	2.80(2–3)	8.67(3–12)	6.17(2–12)	11.17(8–13)	5.67(4–8)	13.33(12–15)	11.67(11–12)	14.67(12–16)	9.67(8–11)
Cici	1.00(0–2)	4.67(3–7)	3.67(3–5)	4.50(4–5)	3.33(2–4)	7.33(7–8)	6.00(5–7)	5.83(4–7)	7.00(6–8)
				Unscripted social-communication					
Anan	0.20(0–1)	2.67(2–3)	1.17(0–3)	3.50(2–5)	1.33(1–2)	3.33(3–4)	2.67(2–3)	3.83(3–5)	3.33(3–4)
Bobo	0.20(0–1)	3.50(1–5)	2.50(1–4)	4.33(3–6)	2.67(2–4)	4.00(3–5)	3.33(3–4)	4.50(3–5)	3.67(3–4)
Cici	0.00(0–0)	1.83(0–3)	2.50(1–3)	2.33(2–3)	0.33(0–1)	4.00(3–5)	4.67(4–5)	4.00(3–5)	4.33(3–5)

Note: VM = video modelling.

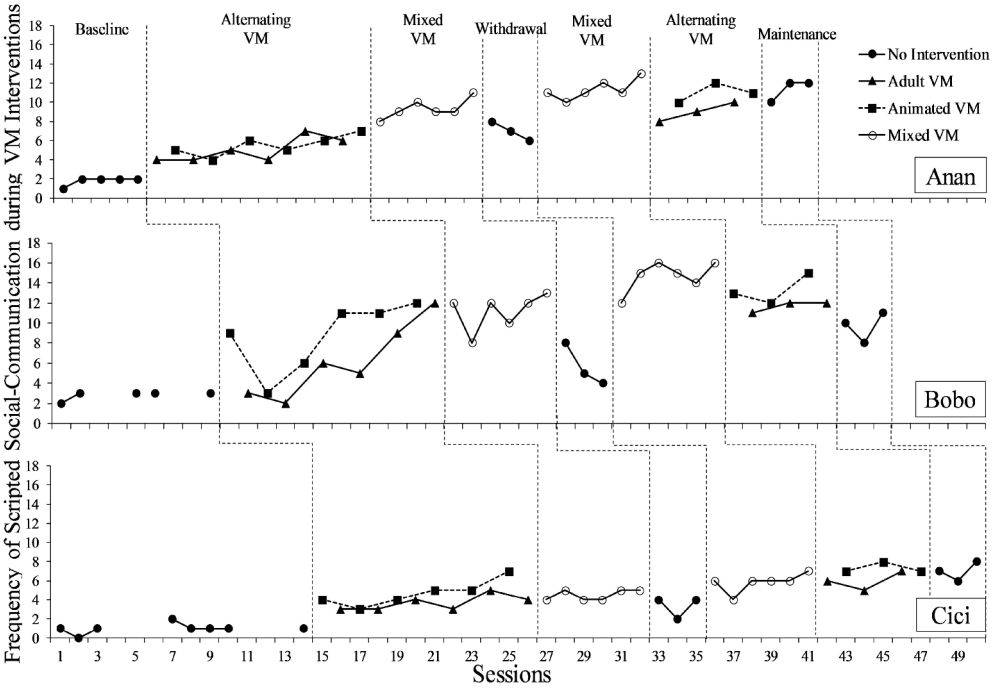


Figure 5. Frequency of scripted social-communication during VM Interventions. Note. VM = video modelling

Scripted Communication

For all children, the frequency of scripted communication gradually increased to a high level with a gradually increasing trend at the beginning of the intervention conditions (see Figure 5). The visual analysis showed no overlap between the baseline and the intervention conditions or between the baseline and maintenance conditions for Anan and Cici. Bobo's data also demonstrated the same pattern, except the initial three data points in the first alternating VM phases, i.e. there was a 10% overlap between baseline and intervention for Bobo.

Comparison of Animated VM and Adult VM

Figure 5 and Table 4 also reveal that all children demonstrated a higher level of scripted social-communication frequency in animated VM compared to the adjacent adult VM conditions. Specifically, we found that the children showed a higher frequency of scripted social communication in response to the first animated VM compared to the first adult VM, and this pattern was repeated in the second animated VM compared to the second adult VM. The Tau-BC results were consistent with the visual analysis.

Comparison of Mixed VM and Single VM

Anan and Bobo's level of scripted communication frequency in the mixed VM conditions was higher than that in the adjacent adult VM and animated VM conditions. Although Cici's level of scripted communication frequency in the mixed VM conditions was lower than that of the

animated VM and adult VM conditions, the Tau-BC results reveal that mixed VM produces higher effects in scripted communication than animated or adult VM for all children.

During the withdrawal and maintenance conditions, the scripted communication frequency data also demonstrated a similar pattern as overall social-communication data. This differentiated result might also reveal that the video viewing was responsible for maintenance.

Unscripted Communication

For all children, the frequency of unscripted communication in most baseline sessions was 0, and the frequency of unscripted communication gradually increased to a high level with a gradual accelerating trend and high variability at the beginning of the intervention conditions (see Figure 6). The visual analysis showed no overlap between the baseline and maintenance conditions for all children, and there was a 6.67%, 6.67%, and 3.33% overlap between baseline and intervention for Anan, Bobo, and Cici, respectively.

Comparison of Animated VM and Adult VM

Anan's and Bobo's level of unscripted communication frequency in the animated VM conditions was higher than that in adjacent adult VM conditions. However, Cici's level of unscripted communication frequency in the animated VM conditions was lower than that in the adjacent adult VM conditions. The Tau-BC results were consistent with the visual analysis.

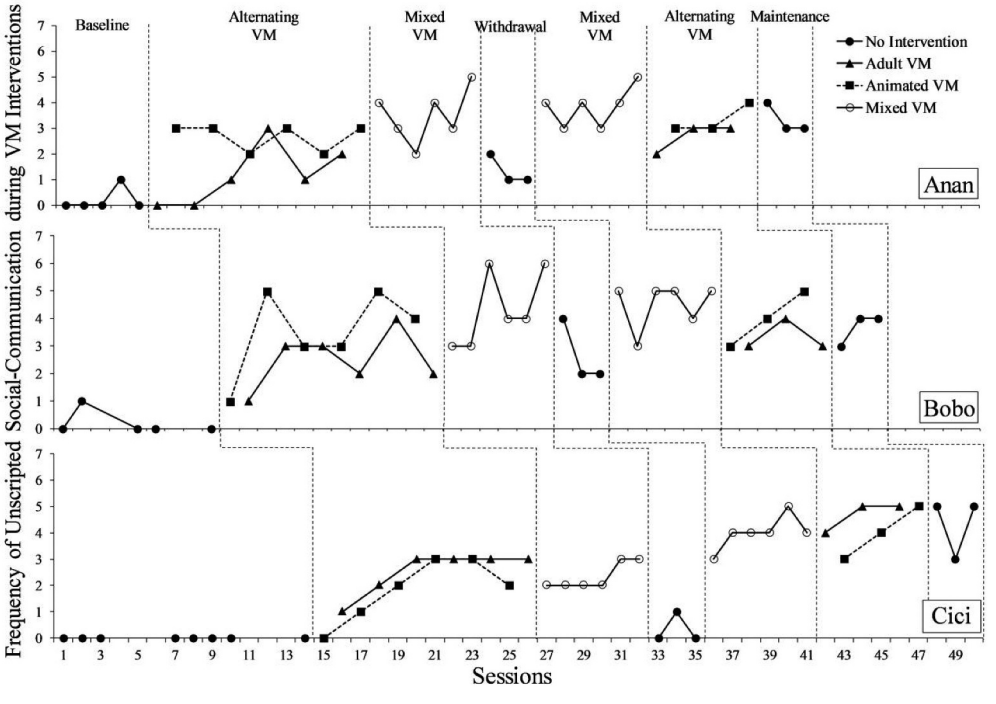


Figure 6. Frequency of unscripted social-communication during VM Interventions. Note. VM = video modelling

Comparison of Mixed VM and Single VM

Anan's and Bobo's level of unscripted communication frequency in the mixed VM conditions was higher than that of adjacent adult VM and animated VM conditions. Cici's level of unscripted communication frequency in the mixed VM conditions was lower than that of adult VM conditions alone. The Tau-BC results were consistent with the visual analysis.

During the withdrawal and maintenance conditions, the data of the unscripted communication frequency also demonstrated a similar pattern as the data of overall social communication. This differentiated result might also reveal that the video viewing was responsible for maintenance.

Discussion

This study was the first attempt to compare the effectiveness of animated VM, adult VM, and mixed VM on social-communication skills for preschoolers with ASD in China. Our results extended previous findings by demonstrating for the first time that children's improvement in social-communication behaviours was greater in response to animated VM than to adult VM. Furthermore, mixed VM was more effective than animated VM or adult VM alone in improving the scripted, unscripted, and overall social communication for two of the three children.

Our study found that all children demonstrated greater improvement in overall social-communication and scripted social-communication behaviours in the animated VM conditions than in the adult VM conditions. This result is consistent with the findings of neuroscience studies that individuals with ASD extract information from cartoon faces more efficiently and faster than from human faces (Grelotti et al., 2005) and adds support for the hypothesis of an exaggerated interest in animated stimuli for individuals with ASD (Dichter et al., 2012). Ohtake (2015) suggested that watching animated videos might be a reinforcing activity for children with ASD, which in turn would promote children's learning. In our study, all children indicated high levels of interest in the animated videos, and they frequently gazed at the animated videos with a smile and verbally expressed that they liked the bunnies. Thus, it appeared that animated VM was the preferable VM for all children in this study compared to adult VM.

Despite the effectiveness of animated VM in teaching children with ASD, to date, the number of studies on animated VM is still very limited due to the limited information on producing animated videos in a simple and economical way. However, with the development of computer technology, it is possible, without too much additional time and knowledge, to produce animated videos with the help of apps. It cannot be emphasised enough that all animated videos were produced with the free and commonly available app, FaceU, and the only step in making an animated video was to convert a human face into an animated face with one click. That is, in 10 s, we can produce a 10-s animated video. This simple and economical way of producing animated videos should greatly encourage practitioners to use this effective strategy in teaching social-communication skills to students with ASD.

Importantly, the unscripted communication of all children improved significantly, consistent with the findings of previous studies that VM can improve the unscripted communication of children with ASD (MacDonald et al., 2009; Maione & Mirenda, 2006). Previous studies have suggested that providing a sufficient range of examples of the

target behaviours seemed to be particularly applicable to promote generative unscripted communication (Maione & Mirenda, 2006). In this study, we provided 10 videos with 10 different LEGO® models for each target script to promote a generalised effect with different activity materials. As a result, all children demonstrated improvement in expressing more generative unscripted communication. However, the comparison of the effectiveness of animated VM and adult VM on unscripted communication yielded inconsistent results across children. Specifically, Anan and Bobo demonstrated a higher frequency of unscripted communication in the animated VM conditions than in the adult VM conditions, whereas Cici demonstrated a higher frequency of unscripted communication in the adult VM conditions. Cici's clinical reports demonstrated excessive adherence to routines, and thus, Cici's most common unscripted communication was 'I want the scarecrow', which usually occurred during the adult VM conditions. We assumed that Cici was so interested in the animated videos that she was very engaged in imitating the scripts; however, Cici was less interested in the videos with adult demonstrations and thus less involved in imitation and more likely to express her preference for the scarecrow model. Given this, we suggest that both the adult VM and the animated VM were effective for unscripted communication, but the level of effectiveness of the two methods was affected by the children's characteristics.

In addition, this study found that mixed VM led to the greatest improvement in all three social-communication outcomes for Anan and Bobo but did not lead to the greatest improvement for Cici. A potential explanation for the relatively low effectiveness of mixed VM for Cici might be her poor observation and attention in complex social environments. Researchers have reported that children with ASD might find it difficult to dynamically adjust their attentional focus and switch the locus of their attention in videos that include more complex social cues because their perceptual defects might make it difficult for them to filter noise signals from complex environments (Chen et al., 2016). In this study, Cici had the lowest cognitive functioning in all children, and her clinical report demonstrated her poor concentration. Therefore, watching two types of videos in the mixed VM might have been challenging for Cici to understand and master the key information. Thus, we suggest that the choice of VM types should be based on individual characteristics.

Limitations

Although the findings from the present study provide preliminary evidence of the higher effectiveness of animated VM than adult VM in teaching social-communication skills to preschoolers with ASD, several limitations should be taken into account when considering the results. First, since only three preschoolers were invited in this single case experimental study, larger and more rigorous experimental designs should be used to ascertain the generalisability of these results. Second, the poor stability of the data of unscripted communication might be a potential threat to internal validity. In this study, the unscripted communication included both reasonable expressions outside the scripts and generalised communication based on the target scripts. The reasonable expressions outside the scripts were usually spontaneous expressions delivered in the participants' own language form; thus, it has a certain contingency. For example, Cici sometimes actively expressed her preference for the scarecrow model. In addition, researchers have long observed that children with ASD are both slower and less consistent in

generalisation (de Marchena et al., 2015). Thus, given the nature of the unscripted communication, it was prone to variability. Third, although all children demonstrated different levels of maintenance of the target behaviours after one month and we provided four LEGO® models in each assessment to promote a generalised effect in different activity materials, we did not carry out a probe to examine generalisation across settings and people.

Directions for Future Research

In this study, we chose animated bunny faces to replace human faces in animated VM and found that animated VM was more effective than adult VM. Previous research also found that preference-enhanced communication interventions facilitated the development of spontaneous and socially oriented communication (Van Der Meer et al., 2014). Thus, future studies could examine whether videos with animated characters that children prefer could enhance the effectiveness of animated VM.

In addition, it is also necessary to further study whether children who receive these VM interventions demonstrate longer-term effects on their social-communication behaviours in more natural contexts towards their peers and family members.

Disclosure Statement

No potential conflict of interest was reported by the author(s).

Funding

This work was funded by the Key project of Chongqing Language Committee [grant number yyk22101]; Research Project on Vocational Education Teaching Reform of Chongqing [grant number Z233527X].

ORCID

Huan Li http://orcid.org/0000-0001-7332-6122

Ethical Approval

All procedures performed in studies involving human participants were in accordance with the ethical standards of the institutional and/or national research committee and with the 1964 Helsinki declaration and its later amendments or comparable ethical standards.

Informed Consent

Parental informed consent was obtained for all individual participants included in the study. Verbal child assent was also obtained from all participants.

References

Ayres, K., & Gast, D. L. (2010). Dependent measures and measurement procedures. In J. R. Ledford & D. L. Gast (Eds.), *Single subject research methodology in behavioral sciences* (pp. 329–381). Routledge.

Bargh, J. A., & Chartrand, T. L. (2000). The mind in the middle: A practical guide to priming and automaticity research. In H. T. Reis & C. M. Judd (Eds.), *Handbook of research methods in social and personality psychology* (pp. 253–285). Cambridge University Press.

Bellini, S., & Akullian, J. (2007). A meta-analysis of video modeling and video self-modeling interventions for children and adolescents with autism spectrum disorders. *Exceptional Children*, *73*(3), 264–287. https://doi.org/10.1177/001440290707300301

Boudreau, E., & D'Entremont, B. (2010). Improving the pretend play skills of preschoolers with autism spectrum disorders: The effects of video modeling. *Journal of Developmental and Physical Disabilities*, *22*(4), 415–431. https://doi.org/10.1007/s10882-010-9201-5

Charlop-Christy, M. H., Le, L., & Freeman, K. A. (2000). A comparison of video modeling with in vivo modeling for teaching children with autism. *Journal of Autism and Developmental Disorders*, *30*(6), 537–552. https://doi.org/10.1023/A:1005635326276

Charlton, C. T., Kellems, R. O., Black, B., Bussey, H. C., Ferguson, R., Goncalves, B., Jensen, M., & Vallejo, S. (2020). Effectiveness of avatar-delivered instruction on social initiations by children with autism spectrum disorder. *Research in Autism Spectrum Disorders*, *71*, 71. https://doi.org/10.1016/j.rasd.2019.101494

Chen, C. H., Lee, I. J., & Lin, L. Y. (2016). Augmented reality-based video-modeling storybook of nonverbal facial cues for children with autism spectrum disorder to improve their perceptions and judgments of facial expressions and emotions. *Computers in Human Behavior*, *55*, 477–485. https://doi.org/10.1016/j.chb.2015.09.033

Corbett, B. A., & Abdullah, M. (2005). Video modeling: Why does it work for children with autism? *Journal of Early and Intensive Behavior Intervention*, *2*(1), 2–8. https://doi.org/10.1037/h0100294

de Marchena, A. B., Eigsti, I. M., & Yerys, B. E. (2015). Brief report: Generalization weaknesses in verbally fluent children and adolescents with autism spectrum disorder. *Journal of Autism and Developmental Disorders*, *45*(10), 3370–3376. https://doi.org/10.1007/s10803-015-2478-6

Dichter, G. S., Felder, J. N., Green, S. R., Rittenberg, A. M., Sasson, N. J., & Bodfish, J. W. (2012). Reward circuitry function in autism spectrum disorders. *Social Cognitive and Affective Neuroscience*, *7*(2), 160–172. https://doi.org/10.1093/scan/nsq095

Drysdale, B., Lee, C. Y. Q., Anderson, A., & Moore, D. W. (2015). Using video modeling incorporating animation to teach toileting to two children with autism spectrum disorder. *Journal of Developmental and Physical Disabilities*, *27*(2), 149–165. https://doi.org/10.1007/s10882-014-9405-1

Ganz, J. B. (2007). Using visual script interventions to address communication skills. *Teaching Exceptional Children*, *40*(2), 54–58. https://doi.org/10.1177/004005990704000207

Gast, D. L. (2010). *Single subject research methodology in behavioral sciences*. Routledge. https://doi.org/10.1177/1088357610391087

Grelotti, D. J., Klin, A. J., Gauthier, I., Skudlarski, P., Cohen, D. J., Gore, J. C., Volkmar, F. R., & Schultz, R. T. (2005). fMRI activation of the fusiform gyrus and amygdala to cartoon characters but not to faces in a boy with autism. *Neuropsychologia*, *43*(3), 373–385. https://doi.org/10.1016/j.neuropsychologia.2004.06.015

Ho, T. Q., Gadke, D. L., Henington, C., Evans McCleon, T. N., & Justice, C. A. (2019). The effects of animated video modeling on joint attention and social engagement in children with autism spectrum disorder. *Research in Autism Spectrum Disorders*, *58*, 83–95. https://doi.org/10.1016/j.rasd.2018.09.004

Howlin, P., Goode, S., Hutton, J., & Rutter, M. (2004). Adult outcome for children with autism. *Journal of Child Psychology and Psychiatry and Allied Disciplines*, *45*(2), 212–229. https://doi.org/10.1111/j.1469-7610.2004.00215.x

Hume, K., Steinbrenner, J. R., Odom, S. L., Morin, K. L., Nowell, S. W., Tomaszewski, B., Szendrey, S., McIntyre, N. S., YücesoyÖzkan, S., & Savage, M. N. (2021). Evidence-based practices for children,

youth, and young adults with autism: Third generation review. *Journal of Autism & Developmental Disorders*, *51*(11), 4013–4032. https://doi.org/10.1007/s10803-020-04844-2

Ingersoll, B. (2011). Recent advances in early identification and treatment of autism. *Current Directions in Psychological Science*, *20*(5), 335–339. https://doi.org/10.1177/0963721411418470

Ingersoll, B., & Wainer, A. (2013). Initial efficacy of project ImPACT: A parent-mediated social communication intervention for young children with ASD. *Journal of Autism and Developmental Disorders*, *43*(12), 2943–2952. https://doi.org/10.1007/s10803-013-1840-9

Koegel, L. K., Ashbaugh, K., Koegel, R. L., Detar, W. J., & Regester, A. (2013). Increasing socialization in adults with asperger's syndrome. *Psychology in the Schools*, *50*(9), 899–909. https://doi.org/10.1002/pits.21715

Krantz, P. J., & McClannahan, L. E. (1998). Social interaction skills for children with autism: A script-fading procedure for beginning readers. *Journal of Applied Behavior Analysis*, *31*(2), 191–202. https://doi.org/10.1901/jaba.1998.31-191

Ledford, J. R., & Gast, D. L.(2018).*Single case research methodology: application in special education and behavioral sciences*. J. R. Ledford & D. L. Gast, (Eds.). (3rd ed.) Routledge.

MacDonald, R., Sacramone, S., Mansfield, R., Wiltz, K., & Ahearn, W. H. (2009). Using video modeling to teach reciprocal pretend play to children with autism. *Journal of Applied Behavior Analysis*, *42*(1), 43–55. https://doi.org/10.1901/jaba.2009.42-43

Maione, L., & Mirenda, P. (2006). Effects of video modeling and video feedback on peer-directed social language skills of a child with autism. *Journal of Positive Behavior Interventions*, *8*(2), 106–118. https://doi.org/10.1177/10983007060080020201

Marcus, A., & Wilder, D. A. (2009). A comparison of peer video modeling and self video modeling to teach textual responses in children with autism. *Journal of Applied Behavior Analysis*, *42*(2), 335–341. https://doi.org/10.1901/jaba.2009.42-335

Mason, R. A., Davis, H. S., Boles, M. B., & Goodwyn, F. (2013). Efficacy of point-of-view video modeling: A meta-analysis. *Remedial and Special Education*, *34*(6), 333–345. https://doi.org/10.1177/0741932513486298

Mason, R. A., Ganz, J. B., Parker, R. I., Burke, M. D., & Camargo, S. P. (2012). Moderating factors of video-modeling with other as model: A meta-analysis of single-case studies. *Research in Developmental Disabilities*, *33*(4), 1076–1086. https://doi.org/10.1016/j.ridd.2012.01.016

Mccoy, K., & Hermansen, E. (2007). Video modeling for individuals with autism: A review of model types and effects. *Education & Treatment of Children*, *30*(4), 183–213. https://doi.org/10.1353/etc.2007.0029

McCoy, A., Holloway, J., Healy, O., Rispoli, M., & Neely, L. (2016). A systematic review and evaluation of video modeling, role-play and computer-based instruction as social skills interventions for children and adolescents with high-functioning autism. *Review Journal of Autism and Developmental Disorders*, *3*(1), 48–67. https://doi.org/10.1007/s40489-015-0065-6

Nikopoulos, C. K., & Keenan, M. (2003). Promoting social initiation in children with autism using video modeling. *Behavioral Interventions*, *18*(2), 87–108. https://doi.org/10.1002/bin.129

Ohtake, Y. (2015). Using a hero as a model in video instruction to improve the daily living skills of an elementary-aged student with autism spectrum disorder: A pilot study. *International Journal of Disability, Development and Education*, *62*(4), 363–378. https://doi.org/10.1080/1034912X.2015.1027179

Ohtake, Y., Takahashi, A., & Watanabe, K. (2015). Using an animated cartoon hero in video instruction to improve bathroom-related skills of a student with autism spectrum disorder. *Education and Training in Autism and Developmental Disabilities*, *50*(3), 343–355.

Qi, C. H., Barton, E. E., Collier, M., & Lin, Y. L. (2018). A systematic review of single-case research studies on using video modeling interventions to improve social communication skills for individuals with autism spectrum disorder. *Focus on Autism and Other Developmental Disabilities*, *33*(4), 249–257. https://doi.org/10.1177/1088357617741282

Rosset, D. B., Rondan, C., Da Fonseca, D., Santos, A., Assouline, B., & Deruelle, C. (2008). Typical emotion processing for cartoon but not for real faces in children with autistic spectrum disorders. *Journal of Autism and Developmental Disorders*, *38*(5), 919–925. https://doi.org/10.1007/s10803-007-0465-2

Rusch, F. R., & Kazdin, A. E. (1981). Toward a methodology of withdrawal designs for the assessment of response maintenance. *Journal of Applied Behavior Analysis*, *14*(2), 131–140. https://doi.org/10.1901/jaba.1981.14-131

Sherer, M., Pierce, K. L., Paredes, S., Kisacky, K. L., Ingersoll, B., & Schreibman, L. (2001). Enhancing conversation skills in children with autism via video technology: Which is better, "self" or "other" as a model? *Behavior Modification*, *25*(1), 140–158. https://doi.org/10.1177/0145445501251008

Shipley-Benamou, R., Lutzker, J. R., & Taubman, M. (2002). Teaching daily living skills to children with autism through instructional video modeling. *Journal of Positive Behavior Interventions*, *4*(3), 166–177. https://doi.org/10.1177/10983007020040030501

Shukla-Mehta, S., Miller, T., & Callahan, K. J. (2010). Evaluating the effectiveness of video instruction on social and communication skills training for children with autism spectrum disorders: A review of the literature. *Focus on Autism and Other Developmental Disabilities*, *25*(1), 23–36. https://doi.org/10.1177/1088357609352901

Silva, C., Da Fonseca, D., Esteves, F., & Deruelle, C. (2015). Motivational approach and avoidance in autism spectrum disorder: A comparison between real photographs and cartoons. *Research in Autism Spectrum Disorders*, *17*, 13–24. https://doi.org/10.1016/j.rasd.2015.05.004

Stone, W. L., Ousley, O. Y., Yoder, P. J., Hogan, K. L., & Hepburn, S. L. (1997). Nonverbal communication in two- and three-year-old children with autism. *Journal of Autism and Developmental Disorders*, *27*(6), 677–696. https://doi.org/10.1023/A:1025854816091

Tarlow, K. R. (2017). An improved rank correlation effect size statistic for single-case designs: Baseline corrected tau. *Behavior Modification*, *41*(4), 427–467. https://doi.org/10.1177/0145445516676750

Topuz, C., & Ulke-Kurkcuoglu, B. (2019). Increasing verbal interaction in children with autism spectrum disorders using audio script procedure. *Journal of Autism and Developmental Disorders*, *49*(12), 4847–4861. https://doi.org/10.1007/s10803-019-04203-w

Van Der Meer, L., Sigafoos, J., Sutherland, D., McLay, L., Lang, R., Lancioni, G. E., O'Reilly, M. F., & Marschik, P. B. (2014). Preference-enhanced communication intervention and development of social communicative functions in a child with autism spectrum disorder. *Clinical Case Studies*, *13*(3), 282–295. https://doi.org/10.1177/1534650113508221

Wechsler, D. (2012). *Wechsler preschool and primary scale of intelligence* (4th ed.). The Psychological Corporation.

Wetherby, A. M., Watt, N., Morgan, L., & Shumway, S. (2007). Social communication profiles of children with autism spectrum disorders late in the second year of life. *Journal of Autism and Developmental Disorders*, *37*(5), 960–975. https://doi.org/10.1007/s10803-006-0237-4

Wichnick-Gillis, A. M., Vener, S. M., & Poulson, C. L. (2016). The effect of a script-fading procedure on social interactions among young children with autism. *Research in Autism Spectrum Disorders*, *26*, 1–9. https://doi.org/10.1016/j.rasd.2016.03.004

Wolery, M., Gast, D. L., & Ledford, J. R. (2018). Comparative Designs. In J. R. Ledford & D. L. Gast (Eds.), *Single case research methodology: Application in special education and behavioral sciences* (3rd ed. pp. 428–497). Routledge. https://www.routledge.com/Single-Case-Research-Methodology-Applications-in-Special-Education-and/Ledford-Gast/p/book/9781138557130